

Trea Turner — Whole-Field Directional Tendency

uc-pos-016-turner-whole-field-directional-001 · UC #42 · dp_uc42 · Phillies Offense (pos) value stream

Requested and delivered 2026-09-10 · data as of 2026-09-09 · Human DPO: Kellen Short

Status: **READY-CONDITIONAL** (diagnostic-only tier) · spot-verification 33/33 PASS

1 · Verdict, before any explanation

Question	Answer
Is Turner pulling outside-zone pitches too often in 2026, drifting from 2023-2025?	No. 2026 outside-zone pull rate (39%, n=71) is flat-to-lower than pooled 2023-25 (43%, n=240); $z = -0.58$, not distinguishable from noise.
Is there a real directional drift anywhere?	Yes - but the opposite direction, and in-zone, not out-of-zone. In-zone pull rate 40% (2026) vs 48% pooled ($z=-2.52$); oppo 35% vs 30% ($z=+2.04$).
Lock this as a formal KPI, or ship as a diagnostic?	Diagnostic-only, provisional (WF-1, WF-2). Sample sizes on the outside-zone slice are too small to lock a KPI around a null result.

The premise, tested against the data rather than assumed, does not survive contact with the data as stated. The 2026 directional shift is real by the repo's own uncertainty-band convention ($|z| \geq 2.5$ for the in-zone pull-rate move), but it is a smaller-sample, in-zone story, not the outside-zone story the working theory named.

2 · Scope and intake note

Kellen's working theory: Turner is pulling too many outside-zone pitches in 2026 rather than serving them the other way, a drift from 2023-2025 Phillies-era behavior. He asked the data-product-owner to lead the narrative against the data, not confirm the premise. The two files he said he was attaching (his working markdown and a photo of a hand sketch) did not reach this session - checked the connected folders, the upload area, and both repos' data-products directories. His prompt text was detailed enough to build from directly and is treated as the intake document of record. The one place this matters is **sort_rank**: invoked as pre-existing with a known RHB-only bug, but not found anywhere in the repo (only *hit_direction* is). This build ships a new, handedness-correct sort_rank rather than guess at undocumented prior logic (see §4).

3 · Data position

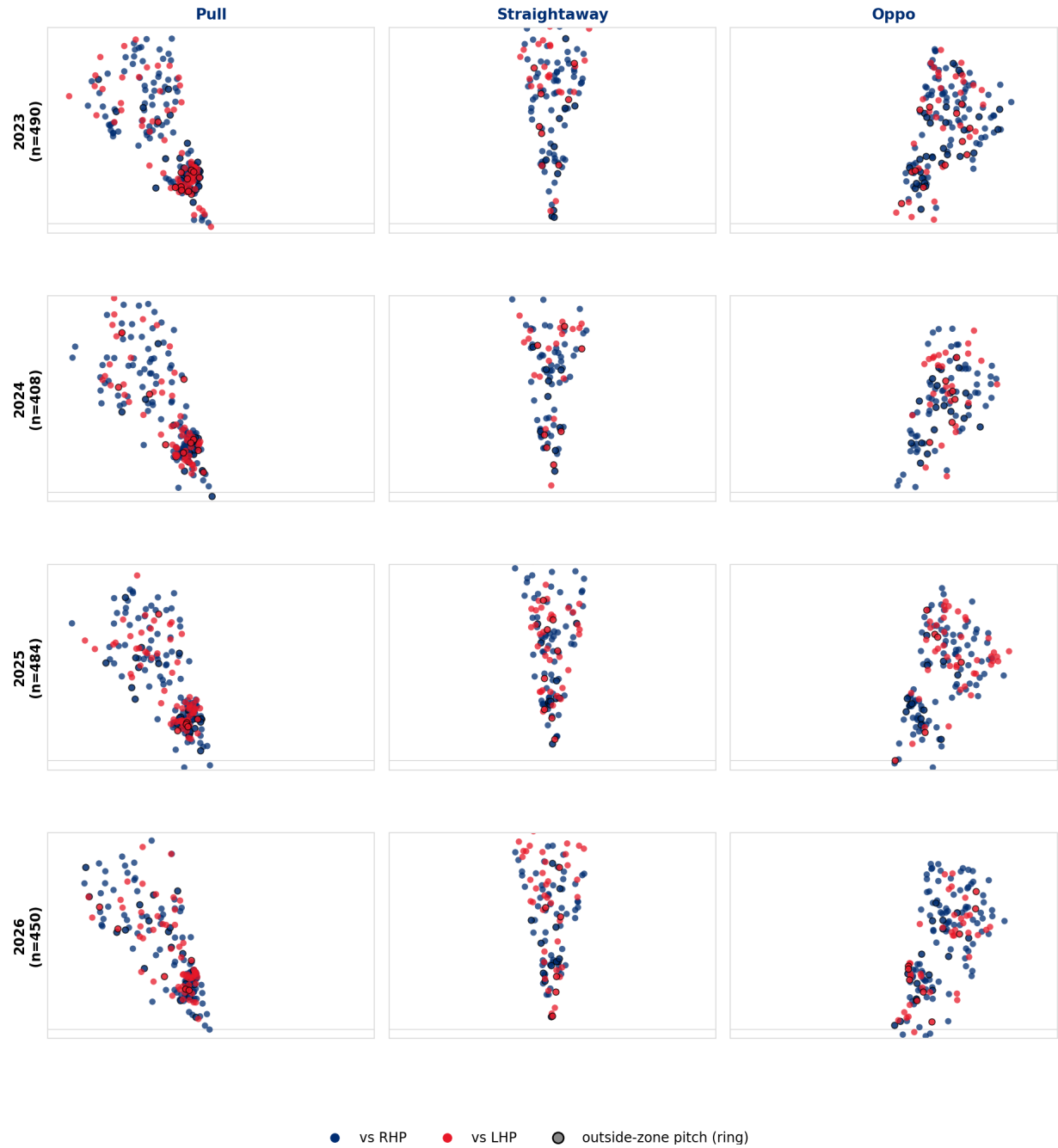
Check	Result
Source	data/phillies/phils_{2023..2026}.parquet - repo governed cache, on-disk contract of get_phillies_data()
Freshness	phils_2026.parquet covers through 2026-09-09
Entity lock	batter == 607208 (confirmed twice before in this repo)
Grain	Pitch-level -> balls in play, Turner only, regular season (game_type=='R')
Coordinate source	hc_x/hc_y -> loc_x/loc_y via governed derive_loc(), inherited verbatim from dp_uc40_kernel.py
hit_direction	Governed, stand-aware +/-4.7-slope classification, verbatim from Baseball Functions.ipynb cell 56
Zone convention	Governed in_zone(): zone<10; NULL-zone rows excluded from both populations

Handedness	Turner is 100% stand=='R' in this dataset; L-branch implemented, not exercised
Volume	1,832 regular-season BIP, 2023-2026 (490 / 408 / 484 / 450)

4 · Whole-field directional map

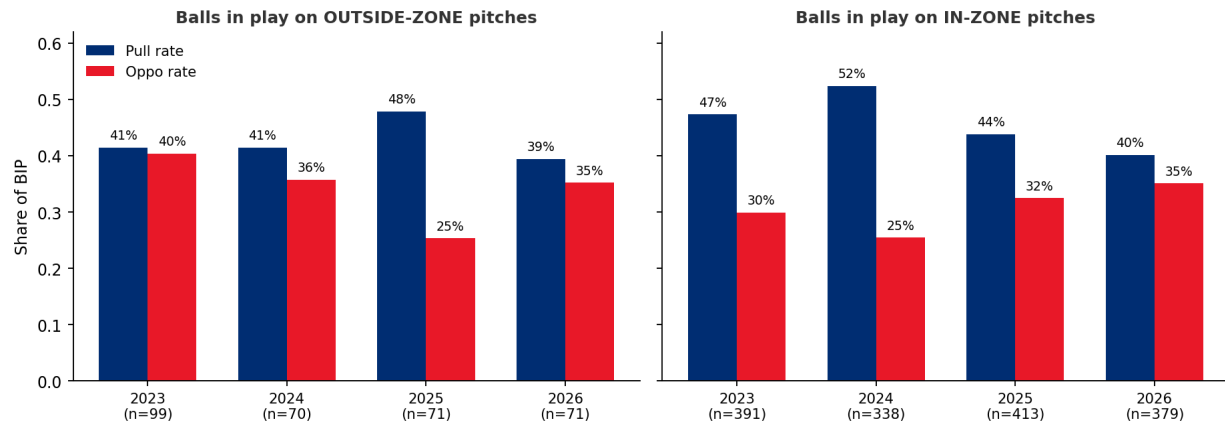
Trea Turner — Whole-Field Directional Tendency, Balls in Play by Season

Facet grid: hit_direction (cols, RHB field-order) x game_year (rows) x p_throws (color) — regular season BIP, 2023-2026



5 · Pull vs. oppo, by season and zone

Trea Turner — Pull vs. Oppo Rate by Season, Split on Pitch Zone (Regular Season BIP)



6 · The finding

Restricting to balls in play off pitches outside the strike zone - the exact population the working theory is about - Turner's 2026 pull rate (39.4%, 28-for-71) is not distinguishable from his pooled 2023-2025 rate (43.3%, 104-for-240; pooled $z = -0.58$) and his oppo rate is essentially flat (35.2% vs 34.6%, $z = +0.10$). There is no outside-zone pull drift in the data, in either direction, at this sample size.

The real movement is on balls in play off pitches **in** the zone: pull rate 40.1% in 2026 vs. 47.5% pooled 2023-2025 ($z = -2.52$) and oppo rate 35.1% vs. 29.5% ($z = +2.04$) - both clear this repo's moderate uncertainty band ($|z| \geq 2.5 / \geq 1.5$, the ST-1 convention). 2026 is the lowest pull-rate, highest oppo-rate season of the four on in-zone contact, and the shift is not monotonic (2024 was his most pull-heavy in-zone season of the window) - 2026 is the outlier year, not the end of a steady trend.

Two honest caveats. First, the outside-zone test is underpowered (71 BIP in 2026, 240 pooled) - a true effect of the hypothesized size could exist without clearing significance at this n . Second, the share of Turner's BIP coming from outside-zone pitches has trended down across the window (20.2% -> 17.2% -> 14.7% -> 15.8%): he is making contact with fewer outside pitches to test the premise on, not more.

Bottom line: the working theory, read literally, is not supported. This organization's falsify-before-describe standard says this ships as a negative finding, not a softened directional hint. The more defensible story is a smaller-sample in-zone shift toward the opposite of the hypothesis - more oppo, less pull - worth tracking with more plate appearances rather than treating as settled.

7 · Open decisions resolved at this layer

1. Static facet grid vs. animated frame (Kellen's call to confirm). Ships the static facet grid - hit_direction (columns) x game_year (rows), colored by p_throws - matching the sketch description and the existing run-spray-chart-60 skill's faceting pattern already in production. An animated-frame-on-season version is a natural follow-on if preferred after seeing this. **Flagged for confirmation, not defaulted.**

2. Formal KPI vs. diagnostic-only. Ships diagnostic-only with two provisional candidates (WF-1 oppo_rate_ooz, WF-2 pull_rate_ooz). Locking a KPI around a metric that just returned a null result, on an under-100-BIP-per-year population, is premature.

3. Ledger ID. uc-pos-016 / UC #42 / dp_uc42 - verified against the ledger index and the data-products/ directory listing at build time; no collision found.

8 · Scope disclosure

Diagnostic-tier build: one subject, one narrow question, a 33-check spot-verification harness rather than an exhaustive independent-recompute harness (c.f. 711 checks on the uc-pos-014 season review). That scope reduction is priced explicitly in the BID, not hidden. Certification: READY-CONDITIONAL.